

VRSAC, Serbia

WMO: 131830

Lat: 45.147N

Lon: 21.310E

Elev: 276

StdP: 14.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.9	17.4	7.3	8.1	14.6	12.7	10.6	19.7	42.8	35.9	38.8	37.4	3.0	40	0.501

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.0	93.5	70.0	90.7	69.9	87.8	68.9	73.4	87.0	71.8	85.6	70.3	83.8	7.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.3	108.9	79.1	67.6	102.4	77.2	66.0	96.8	75.5	37.1	86.9	35.6	85.6	34.4	84.0	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.7	30.1	25.9	DB	2.4	97.3	8.4	4.1	-3.6	100.3	-8.5	102.7	-13.2	105.0	-19.2	107.9	(4)
			WB	2.1	76.3	8.4	2.7	-3.9	78.2	-8.8	79.8	-13.5	81.3	-19.6	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	34.2	38.1	45.3	55.3	63.5	69.3	72.5	73.1	64.4	54.9	46.5	36.6
	DBStd	15.97	9.98	10.24	9.00	7.88	7.19	6.81	6.31	6.80	7.50	8.67	9.58	9.33
	HDD50	1738	496	347	200	37	1	0	0	0	0	52	179	426
	HDD65	4657	955	753	611	302	116	34	9	9	100	328	558	882
	CDD50	3398	6	14	54	196	419	578	697	716	432	203	73	10
	CDD65	840	0	0	1	11	69	161	241	260	81	14	2	0
	CDH74	9278	0	0	2	96	601	1784	2876	3059	789	70	1	0
	CDH80	3605	0	0	0	14	112	640	1241	1358	234	6	0	0
Wind	WSAvg	8.0	8.8	9.4	8.7	9.1	7.2	6.0	5.7	6.4	7.1	8.9	10.7	8.0
Precipitation	PrecAvg	27.0	1.7	1.6	1.6	2.3	2.7	3.2	3.3	2.3	2.5	2.1	1.9	2.0
	PrecMax	37.0	3.3	3.3	4.2	4.8	4.9	7.4	9.7	7.3	7.3	5.0	5.2	5.2
	PrecMin	14.1	0.4	0.0	0.4	0.2	0.9	0.6	0.1	0.1	0.3	0.0	0.0	0.3
	PrecStd	5.7	0.8	0.9	1.0	1.1	1.1	1.7	2.4	1.6	1.5	1.3	1.1	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.9	64.7	72.1	82.1	86.4	93.3	97.4	98.6	91.5	79.9	71.9	60.7
		MCWB	49.5	51.2	54.8	61.7	66.8	71.1	70.6	70.1	67.5	63.2	58.2	51.3
	2%	DB	55.2	59.4	67.9	76.6	82.8	89.8	93.3	94.4	86.0	75.6	67.6	57.3
		MCWB	47.4	49.7	53.2	59.3	65.4	71.0	70.2	69.7	65.9	61.2	56.1	49.4
	5%	DB	52.1	56.1	64.0	72.5	80.4	87.1	90.0	91.0	82.3	72.0	64.1	53.9
		MCWB	45.6	47.5	51.2	57.2	64.3	69.6	70.2	69.5	64.4	59.7	54.3	46.9
	10%	DB	48.7	52.5	60.1	69.1	77.3	83.9	86.9	87.6	78.7	68.9	60.7	50.4
		MCWB	43.4	45.5	49.4	55.4	62.8	68.5	69.2	68.3	63.4	58.7	52.9	44.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.4	52.8	56.5	63.1	69.8	75.8	75.1	75.9	69.6	65.2	59.6	53.7
		MCDB	57.0	62.3	68.5	77.7	81.7	87.2	88.3	88.5	84.8	76.5	68.8	58.2
	2%	WB	48.2	50.3	54.2	60.5	67.4	72.8	73.2	72.8	67.5	62.9	57.0	49.9
		MCDB	54.4	57.9	65.1	73.6	79.2	85.7	86.9	86.9	81.0	72.2	65.4	56.3
	5%	WB	46.1	48.3	52.2	58.6	65.7	70.9	71.7	71.1	65.9	61.1	55.1	47.5
		MCDB	51.4	55.2	62.2	70.0	77.3	83.7	85.9	84.9	78.0	69.8	62.6	53.1
	10%	WB	43.9	46.2	50.2	56.6	63.9	69.3	70.3	69.6	64.4	59.3	53.0	45.3
		MCDB	48.2	51.7	58.4	66.7	75.1	81.4	83.7	83.6	75.6	68.0	60.3	49.9
Mean Daily Temperature Range	5% DB	MDBR	11.1	13.7	17.4	18.3	19.4	20.8	22.2	22.0	19.2	17.4	13.7	10.7
		MCDBR	14.6	18.7	22.7	23.8	23.1	25.1	26.3	26.0	24.6	22.2	17.5	14.2
		MCWBR	9.6	11.3	12.2	11.6	10.9	10.8	10.0	9.3	10.2	11.6	10.0	8.9
	5% WB	MCDBR	13.3	17.4	20.4	21.8	21.5	23.6	22.9	22.8	21.3	20.1	16.1	13.6
MCWBR		9.1	10.7	11.5	11.2	10.7	10.8	9.8	9.8	9.7	11.2	9.6	9.2	
Clear-Sky Solar Irradiance	taub	0.344	0.371	0.406	0.461	0.448	0.441	0.449	0.462	0.435	0.395	0.361	0.342	
	taud	2.320	2.238	2.197	2.057	2.161	2.217	2.194	2.157	2.199	2.301	2.349	2.347	
	Ebn at Noon	237	252	259	253	262	264	260	251	246	240	228	224	
	Edh at Noon	26	33	40	50	46	44	44	45	39	31	25	23	
All-Sky Solar Radiation	RadAvg	382	648	1045	1454	1762	1955	1964	1748	1220	833	502	315	
	RadStd	45	87	132	145	167	178	151	166	132	92	44	46	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.44	N/A	N/A	+2.59	N/A	N/A	N/A	N/A	N/A	N/A
Regional (61 neighbors)		+1.21	N/A	N/A	N/A	+0.68	N/A	-200	-311	+212	+115

Nomenclature: See separate page